

CLOUD ENGINEERING | STUDENT ASSIGNMENT

AWS S3

Document Portal

Secure Document Sharing Platform using Amazon S3, IAM and CloudFormation

Prepared by

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SERVICES

S3, IAM, CloudFormation

ENVIRONMENT

Lab

DATE

September 29, 2026

INTRODUCTION

Project Overview

ABC Training Solutions needs a reliable way to store and share documents in the cloud. As a Junior Cloud Engineer, I designed and deployed a secure document platform on Amazon S3. The infrastructure is created with AWS CloudFormation, protected by versioning, and shared through IAM users and a carefully scoped bucket policy.

Project goals

Deploy the bucket with CloudFormation • Enable versioning • Keep Block Public Access on at first • Give IAM users controlled access • Expose only the public/ folder • Protect private files • Prove that deleted or overwritten files can be recovered.

Architecture summary

Item	Value
Bucket name	abc-training-documents-souleymane-diallo
Region	us-east-1 (N. Virginia)
Tags	Environment = Lab, Owner = Souleymane Diallo
Versioning	Enabled
Folder structure	company-documents/ → public/, shared/, private/
IAM users	employee1 (read/write), contractor1 (read-only, shared/ only)
Stack name	S3-Document-Portal-Stack

PART 1

Create the S3 Bucket with CloudFormation

Objective

Deploy the bucket as infrastructure as code, with versioning on, Public Access Block on, and the required tags.

CloudFormation template (s3-document-portal.yaml)

```
AWSTemplateFormatVersion: "2010-09-09"
Description: ABC Training Solutions - Secure Document Portal

Parameters:
  BucketName:
    Type: String
    Default: abc-training-documents-souleymane-diallo
  OwnerName:
    Type: String
    Default: Souleymane Diallo

Resources:
  DocumentBucket:
    Type: AWS::S3::Bucket
    Properties:
      BucketName: !Ref BucketName
      VersioningConfiguration:
        Status: Enabled
      PublicAccessBlockConfiguration:
        BlockPublicAcls: true
        BlockPublicPolicy: true
        IgnorePublicAcls: true
        RestrictPublicBuckets: true
      Tags:
        - Key: Environment
          Value: Lab
        - Key: Owner
          Value: !Ref OwnerName

Outputs:
  BucketArn:
    Value: !GetAtt DocumentBucket.Arn
```

Deployment steps

1. Open the AWS Console → CloudFormation → Create stack → With new resources.
2. Upload the file s3-document-portal.yaml and click Next.
3. Name the stack S3-Document-Portal-Stack and keep the default parameters.

- Review, acknowledge, and create the stack. Wait for CREATE_COMPLETE.
- In S3, open the bucket and create the folders public/, shared/ and private/.

CLI alternative: `aws cloudformation deploy --template-file s3-document-portal.yaml --stack-name S3-Document-Portal-Stack`

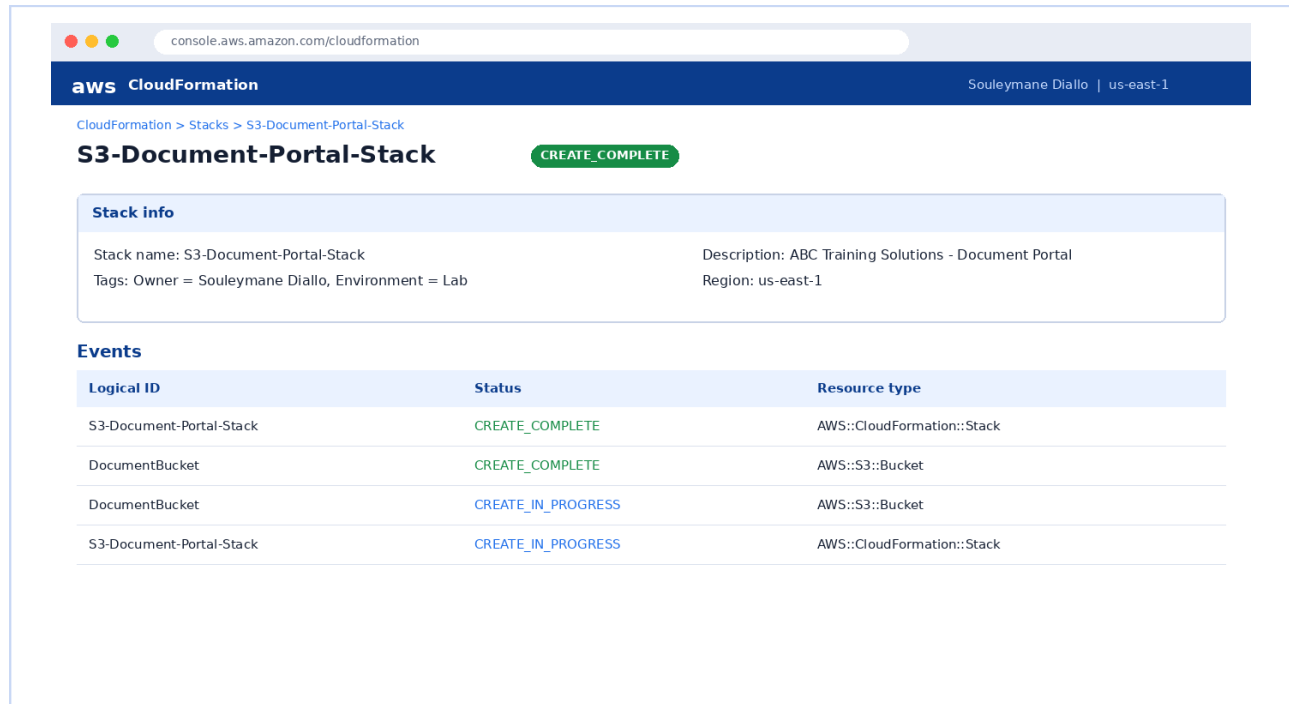


Figure 1: CloudFormation stack showing CREATE_COMPLETE.

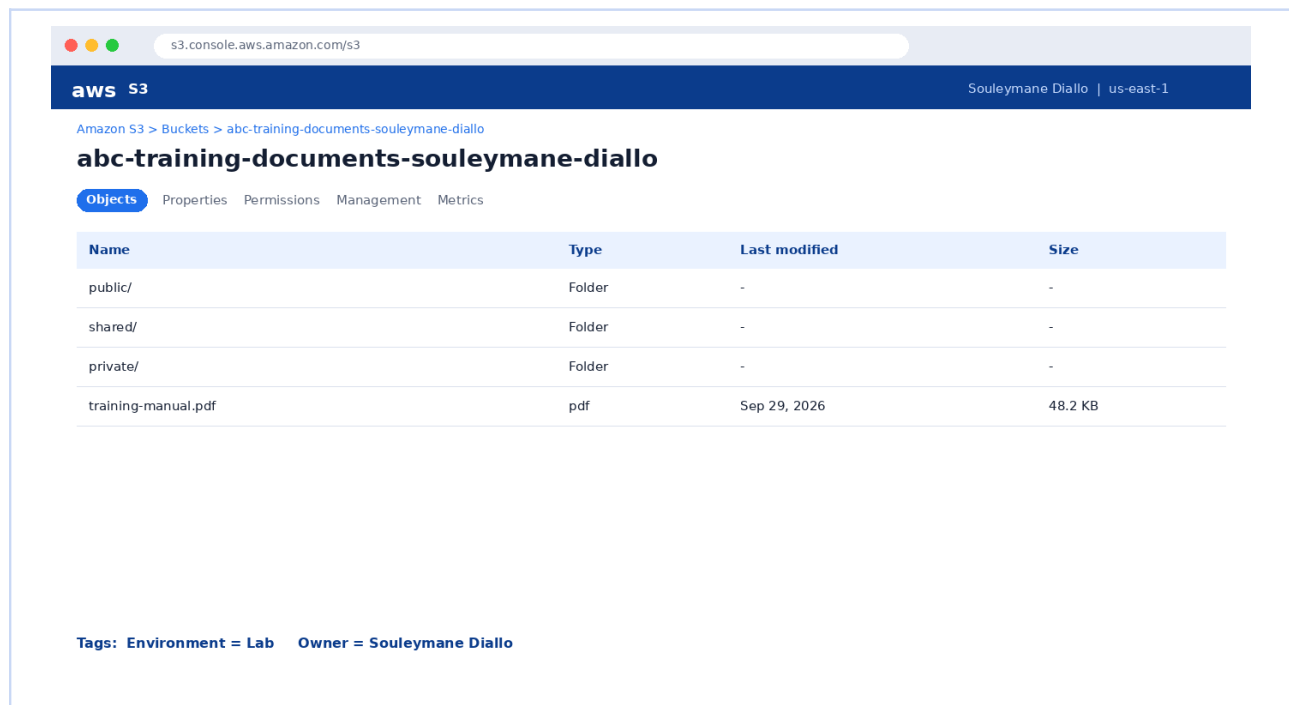
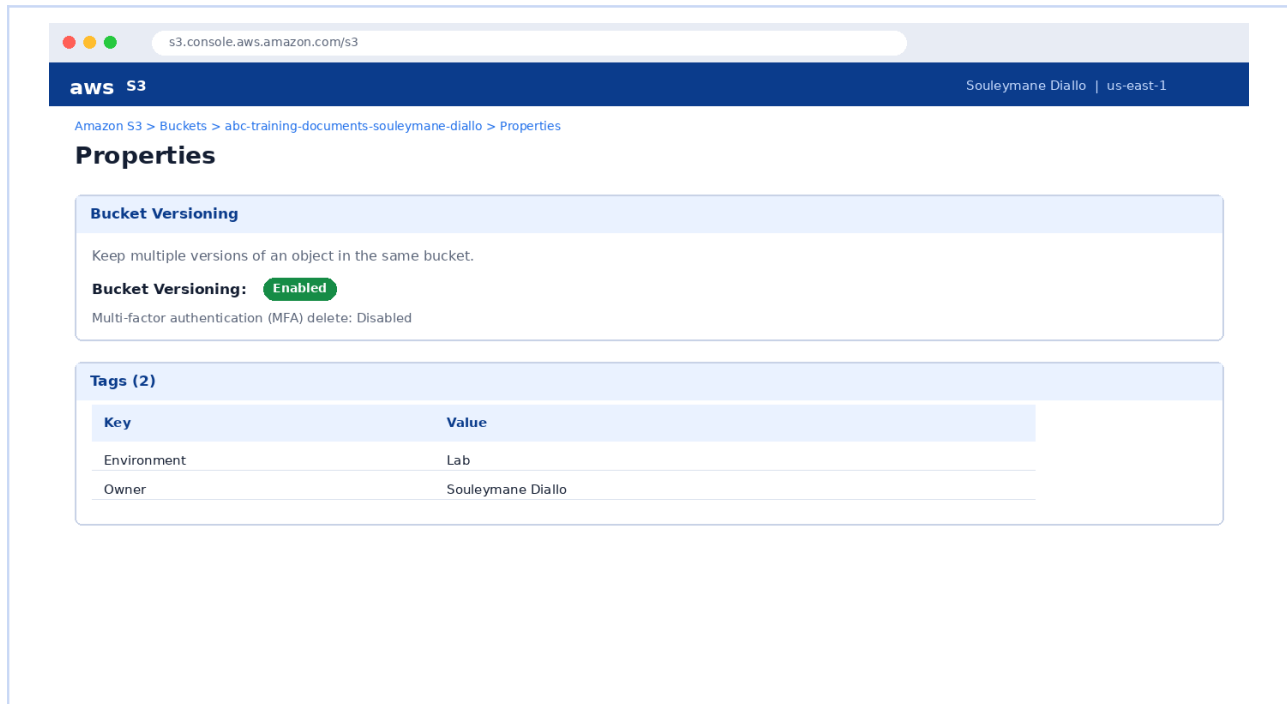


Figure 2: S3 bucket with the public/, shared/ and private/ folders.



s3.console.aws.amazon.com/s3

aws S3 Souleymane Diallo | us-east-1

Amazon S3 > Buckets > abc-training-documents-souleymane-diallo > Properties

Properties

Bucket Versioning

Keep multiple versions of an object in the same bucket.

Bucket Versioning: **Enabled**

Multi-factor authentication (MFA) delete: Disabled

Tags (2)

Key	Value
Environment	Lab
Owner	Souleymane Diallo

Figure 3: Bucket Versioning enabled and required tags applied.

PART 2

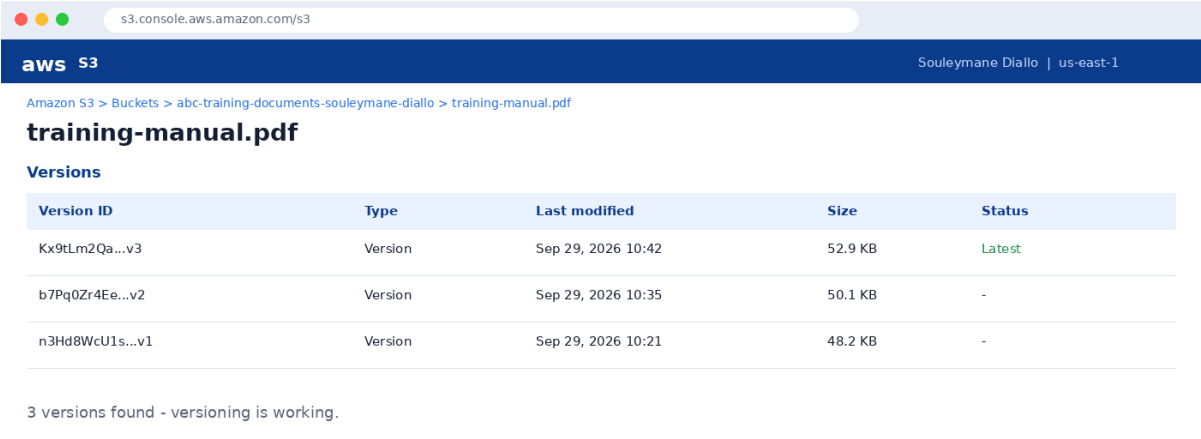
Enable Document Version Control

Objective

Show that S3 keeps every uploaded revision of training-manual.pdf.

1. Upload training-manual.pdf to the bucket (Version 1).
2. Edit the file locally (add a page or change the text) and upload it with the same name (Version 2).
3. Edit it again and upload once more (Version 3).
4. Open the object → Versions tab and confirm that three different Version IDs exist.

```
aws s3 cp training-manual.pdf
s3://abc-training-documents-souleymane-diallo/training-manual.pdf
aws s3api list-object-versions --bucket abc-training-documents-souleymane-diallo --
prefix training-manual.pdf
```



The screenshot shows the AWS S3 console interface. The breadcrumb navigation is "Amazon S3 > Buckets > abc-training-documents-souleymane-diallo > training-manual.pdf". The main heading is "training-manual.pdf". Below it, the "Versions" tab is selected, displaying a table with three versions of the object.

Version ID	Type	Last modified	Size	Status
Kx9tLm2Qa...v3	Version	Sep 29, 2026 10:42	52.9 KB	Latest
b7Pq0Zr4Ee...v2	Version	Sep 29, 2026 10:35	50.1 KB	-
n3Hd8WcU1s...v1	Version	Sep 29, 2026 10:21	48.2 KB	-

3 versions found - versioning is working.

Figure 4: Object Versions: three versions of training-manual.pdf.

PART 3

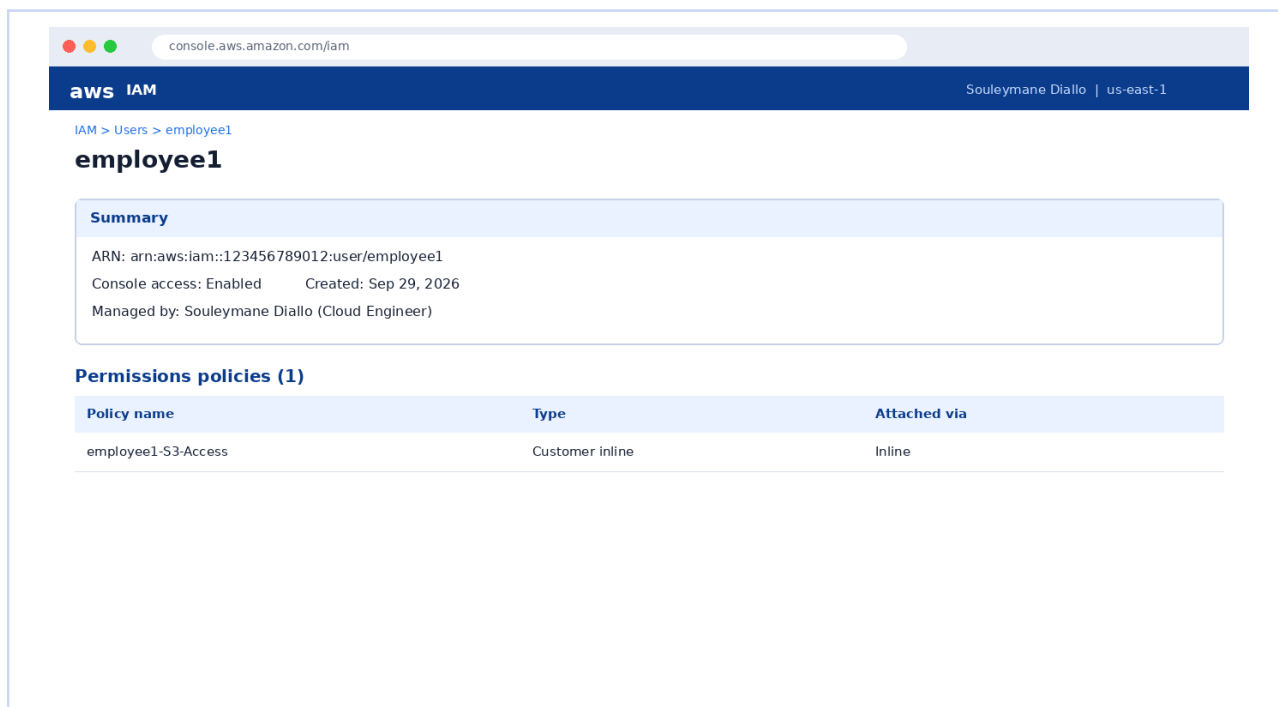
Create the IAM User employee1

Objective

Give an employee the minimum permissions needed: list the bucket, download files and upload files.

1. IAM → Users → Create user, named employee1.
2. Enable console access and set a password (require change at first sign-in).
3. Choose Attach policies directly → Create inline policy → JSON.
4. Paste the policy below and save it as employee1-S3-Access.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::abc-training-documents-souleymane-diallo"
    },
    {
      "Effect": "Allow",
      "Action": ["s3:GetObject", "s3:PutObject"],
      "Resource": "arn:aws:s3:::abc-training-documents-souleymane-diallo/*"
    }
  ]
}
```



The screenshot shows the AWS IAM console interface. At the top, the breadcrumb navigation is 'IAM > Users > employee1'. The user 'employee1' is selected. The 'Summary' section displays the following information: ARN: arn:aws:iam::123456789012:user/employee1, Console access: Enabled, Created: Sep 29, 2026, and Managed by: Souleymane Diallo (Cloud Engineer). Below the summary, the 'Permissions policies (1)' section shows a table with one policy attached.

Policy name	Type	Attached via
employee1-S3-Access	Customer inline	Inline

Figure 5: IAM user employee1 with its permissions policy.

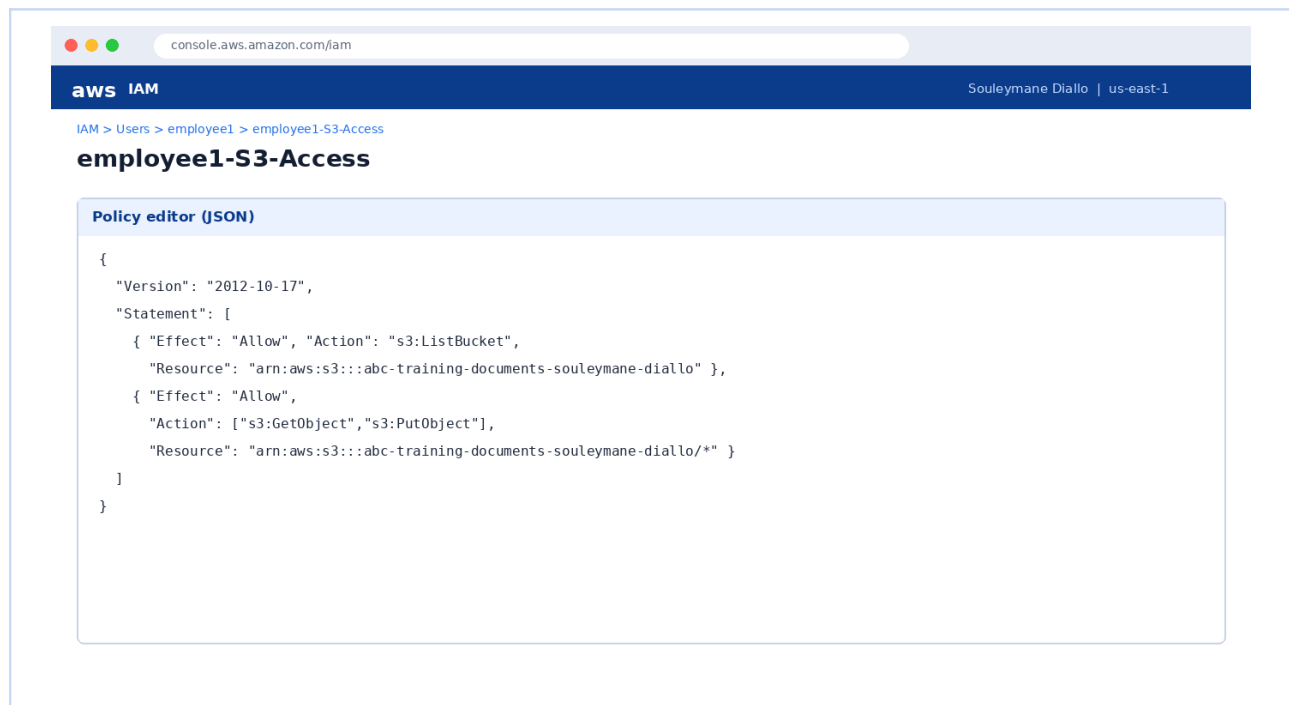


Figure 6: IAM policy granting *ListBucket*, *GetObject* and *PutObject*.

PART 4

Test IAM User Access

Objective

Confirm that employee1 can list, upload and download files, and cannot do anything more.

1. Sign in to the IAM sign-in URL as employee1 (or configure a CLI profile).
2. List the bucket contents.
3. Upload a test file to the shared/ folder.
4. Download training-manual.pdf.
5. Try an action that is not allowed (for example, deleting an object) and note that it is denied.

```
aws s3 ls s3://abc-training-documents-souleymane-diallo/ --profile employee1
aws s3 cp test.txt s3://abc-training-documents-souleymane-diallo/shared/test.txt --
profile employee1
aws s3 cp s3://abc-training-documents-souleymane-diallo/training-manual.pdf . --
profile employee1
```

Result

All three allowed actions succeeded. Actions outside the policy are rejected with "AccessDenied", which proves least-privilege access works.

PART 5

Public Access for Non-IAM Users

Objective

Let anyone on the internet read files in public/ only, while private/ and shared/ stay protected.

1. Bucket → Permissions → Block public access → Edit, and untick Block public access to buckets and objects granted through new public bucket or access point policies (and the matching restrict option). Confirm.
2. Bucket → Permissions → Bucket policy → Edit.
3. Paste the read-only policy below and save.
4. Upload welcome.pdf into public/ and open its Object URL in a private browser window.
5. Try the URL of a file in private/ and verify Access Denied.

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Sid": "PublicReadForPublicFolder",
    "Effect": "Allow",
    "Principal": "*",
    "Action": "s3:GetObject",
    "Resource": "arn:aws:s3:::abc-training-documents-souleymane-diallo/public/*"
  }]
}
```

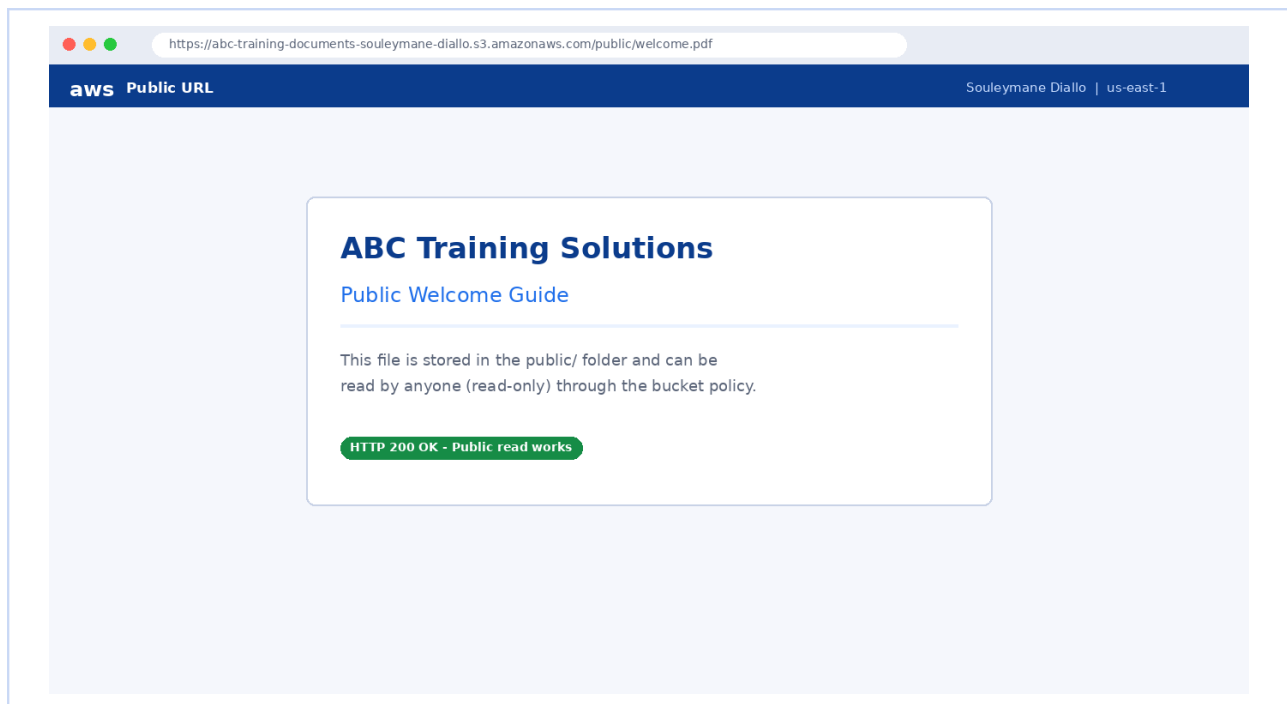


Figure 7: File in public/ opens for anyone (public read works).

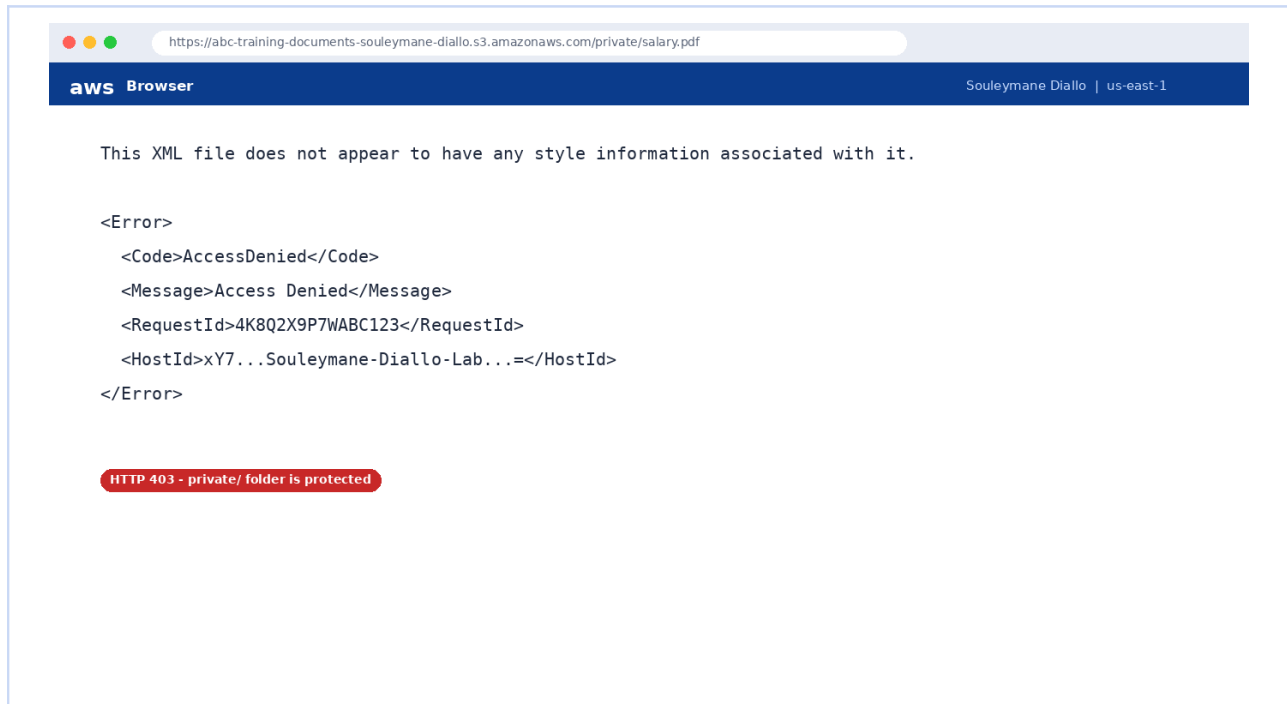


Figure 8: File in `private/` returns Access Denied (403).

Security note

Only the `public/` prefix is exposed. Never store confidential data in that folder, and re-enable Block Public Access when the lab is finished.

PART 6

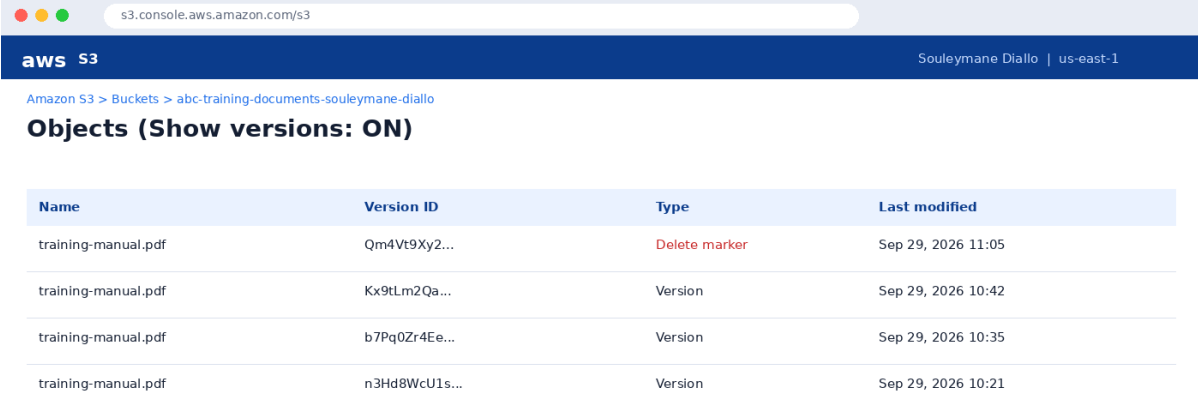
S3 Versioning Recovery Lab

Objective

Recover a file after accidental deletion by removing the delete marker.

1. Select training-manual.pdf and click Delete. Type "delete" to confirm. With versioning on, S3 adds a delete marker instead of erasing data.
2. Switch on Show versions. The file now has a version of type Delete marker at the top.
3. Select only the delete marker and choose Delete, then type "permanently delete".
4. Turn off Show versions. training-manual.pdf is visible again.

```
aws s3api list-object-versions --bucket abc-training-documents-souleymane-diallo --  
prefix training-manual.pdf  
aws s3api delete-object --bucket abc-training-documents-souleymane-diallo --key  
training-manual.pdf --version-id <DELETE_MARKER_VERSION_ID>
```



Name	Version ID	Type	Last modified
training-manual.pdf	Qm4Vt9Xy2...	Delete marker	Sep 29, 2026 11:05
training-manual.pdf	Kx9tLm2Qa...	Version	Sep 29, 2026 10:42
training-manual.pdf	b7Pq0Zr4Ee...	Version	Sep 29, 2026 10:35
training-manual.pdf	n3Hd8WcU1s...	Version	Sep 29, 2026 10:21

The delete marker hides the file. Older versions still exist.

Figure 9: Delete marker shown above the earlier versions.

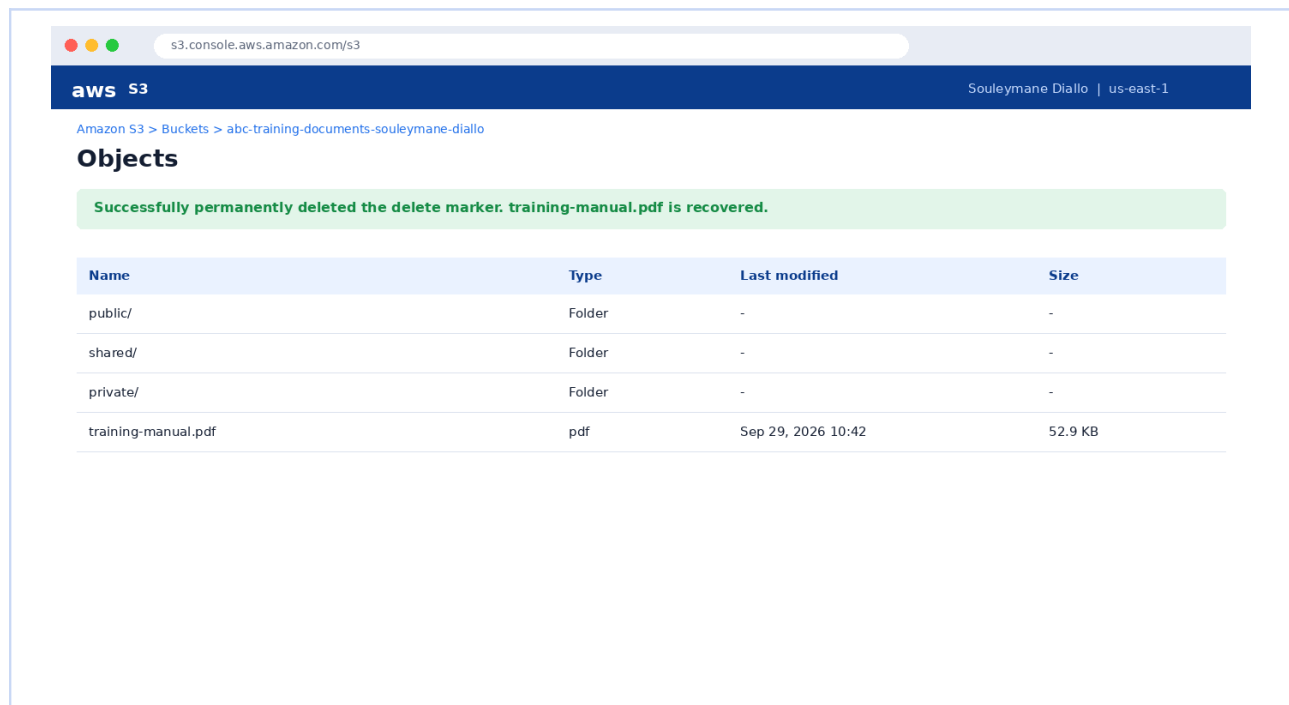


Figure 10: Recovery: delete marker removed and the file is back.

PART 7

Restore a Previous Version

Objective

Roll a file back to Version 1 after it was overwritten by later edits.

1. Upload three different versions of the same file (already done in Part 2).
2. Open the object → Versions tab and select Version 1.
3. Download Version 1, then upload it again under the same key. It becomes the newest version.
4. Alternatively, copy the old version onto itself with the CLI command below.

```
aws s3api copy-object --bucket abc-training-documents-souleymane-diallo \
  --key training-manual.pdf \
  --copy-source "abc-training-documents-souleymane-diallo/training-manual.pdf?
versionId=<VERSION_1_ID>"
```

Why this works

S3 never overwrites old data when versioning is on. Restoring simply promotes an older version to become the current one.

BONUS CHALLENGE**contractor1: Read-Only Access to shared/**

contractor1 may list and read only the shared/ folder. Everything else is implicitly denied.

```
{
  "Version": "2012-10-17",
  "Statement": [
    { "Effect": "Allow",
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::abc-training-documents-souleymane-diallo",
      "Condition": { "StringLike": { "s3:prefix": ["shared/*", "shared/"] } } },
    { "Effect": "Allow",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::abc-training-documents-souleymane-diallo/shared/*" }
  ]
}
```

SUMMARY

Deliverables Checklist

Deliverable	Screenshot	Section
CloudFormation Stack	Figure 1	Part 1
S3 Bucket	Figure 2	Part 1
Versioning Enabled	Figure 3	Part 1
IAM User	Figure 4	Part 3
IAM Policy	Figure 5	Part 3
Public Access	Figure 6	Part 5
Access Denied	Figure 7	Part 5
Object Versions	Figure 8	Part 2
Delete Marker	Figure 9	Part 6
Recovery	Figure 10	Part 6

Learning outcomes achieved

Skill	What I demonstrated
AWS CloudFormation	Built and deployed the bucket as code
Amazon S3	Folders, tags, object management
S3 Versioning	Multiple versions, delete markers, recovery
IAM Users & Policies	Least-privilege users employee1 and contractor1
Bucket Policies	Public read on public/ only
Public vs Private Access	Verified 200 vs 403 responses
Cloud Storage Administration	Secure, recoverable document storage

Conclusion

The ABC Training Solutions document portal is secure, versioned and recoverable. Public content is isolated in one folder, employees and contractors receive only the access they need, and accidental deletions or bad edits can be undone in minutes.

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